

Comparative Analysis of MPC, RL and P&O for MPPT of PV Systems Under Partial Shading Conditions

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Abstract— This paper provides a comparison between three maximum power point tracking (MPPT) methods in photovoltaic (PV) systems: Model Predictive Control (MPC), Reinforcement Learning (RL), and traditional perturb and observe (P&O) methods. The goal of the paper is to compare and analyze how well the techniques perform, their adaptability, and effectiveness in varying operating conditions. MPC uses a model-predictive framework to optimize the control actions over a finite horizon, while reinforcement learning uses an adaptive learning system to maximize power extraction without explicit modeling of the system. The focus of this work is on the distinguishing features of the three types of MPPT, their response time, compute intensity, and adaptability. MPC quickly maximizes and adapts to the MPP while respecting system constraints, making it consist with previously defined dynamics. RL relies on continuous learning and adaption to changes in irradiance and temperature and provides a flexible response to changing environments that have a great deal of uncertainty. RL typically requires a great deal of training and tuning to optimize the algorithm. This paper summarizes the tradeoffs between predictive control and learning algorithms and helps guide researchers and practitioners on the most effective approaches of MPPT for PV systems, resulting in efficient and sustainable PV energy management practices.

Keywords— *Maximum Power Point Tracking, Perturb and Observe, Model Predictive Control, Reinforcement Learning, Photovoltaic Systems, Renewable Energy, System Response.*

I. INTRODUCTION

The growing interest in renewable energy is fueled by the pressing need to cut down on greenhouse gas emissions and address the challenges posed by climate change. [1]. Photovoltaic (PV) technology stands out in the renewable energy landscape for its direct conversion of solar radiation into electrical energy, leading to its widespread adoption. These systems are widely deployed across residential, commercial, and industrial sectors, contributing to a more sustainable energy landscape [2]. Conventional PV systems often struggle to maintain high efficiency because they are affected by changing environmental conditions, such as fluctuations in solar irradiance, variations in temperature, and the impact of shading. These fluctuations can result in considerable energy losses because PV panels, when either fixed in position or managed with less-than-optimal control strategies, struggle to efficiently capture and convert solar energy. [3],[4]. Consequently, enhancing PV energy capture efficiency remains a critical research focus [5].

To maximize energy output from PV systems, Maximum Power Point Tracking (MPPT) techniques are employed, dynamically adjusting operating parameters to ensure optimal power delivery by continuously adjusting the

operating voltage and current to ensure maximum energy extraction [6]. Over time, numerous MPPT techniques have been developed, with traditional methods including Perturb and Observe (P&O) [7], Incremental Conductance (IC) [8], and Fuzzy Logic Control (FLC) [9]. While these methods are widely used, they exhibit limitations in tracking speed, accuracy, and adaptability under rapidly changing environmental conditions. To address these challenges, advanced optimization algorithms such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), and Artificial Neural Networks (ANNs) have been introduced to enhance MPPT accuracy and dynamic response [10]. However, these methods still struggle to achieve optimal performance under non-uniform operating conditions, such as partial shading scenarios [10]. These challenges have encouraged researchers to investigate advanced techniques such as Model Predictive Control (MPC) and Reinforcement Learning (RL), which provide greater flexibility and enable real-time optimization to enhance system performance.

MPC has risen to prominence as an effective control scheme for MPPT due to its ability to predict and improve overall system performance over a defined predictive horizon. MPC uses a predictive mathematical model of the PV plant to predict future states and develop control moves that optimize energy while satisfying operational constraints. However, its effectiveness can be limited by modeling inaccuracies and uncertainties such as shading effects, dust accumulation, and fluctuating loads [11].

Conversely, RL offers a flexible, data-driven method of learning optimal control strategies by automating the learning process through trial and error with the environment. Whereas, MPC relies on a defined model, RL continuously adapts its control strategy based on a system response, which allows an RL-driven method to respond well to non-linearity and an unpredictable behavior of the system. However, training needs, computation, and convergence time affect RL-based MPPT, which may affect the practicality of implementation in real-time systems.

While MPPT methods have improved a lot, it is still a big challenge to maintain optimal tracking efficiency in rapidly changing environmental conditions. The traditional MPPT methods suffer from slow convergence, and in many situations they are inaccurate or high in complexity. The MPC and RL methods are limited to model dependence and inefficient learning, respectively. The goal of this paper is to:

1. To implement and test the MPPT strategies—P&O, MPC, and RL—under variable irradiance scenarios, especially partial shading.

2. Assess the performance of each approach in terms of tracking accuracy, energy yield, and system resilience.

By addressing these challenges, this study seeks to advance PV system optimization, contributing to the development of more efficient, intelligent, and reliable renewable energy solutions.

II. RELATED WORK

By enabling solar panels to follow the sun's position as it changes throughout the day, developments in solar tracking systems have significantly improved the efficiency of PV systems. Although significant for its time, the manual method used by early solar tracking systems to account for the sun's movement was neither particularly effective nor affordable. In the end, automatic solar tracking systems were created, particularly in the 1970s when the global energy crisis sparked interest in finding alternative energy sources, such as solar power. The concept of solar tracking was first proposed at the close of the 1800s. Augustin Mouchot created solar thermal engines in the 1860s, which were the first examples of solar energy being used for non-traditional purposes. But it wasn't until much later that the mechanical complexity needed for solar tracking was fully realized. [12]. In the early 20th century, solar trackers emerged in concentrated solar power (CSP) projects, which required precision to focus sunlight on a small area, typically a receiver or a boiler. These early systems were mechanical and manual, requiring engineers to adjust the angle of solar collectors manually [13]. By the late 20th century, advancements in electronics and computing played a pivotal role in the evolution of solar tracking systems. The advent of precise sensors, microcontrollers, and motors enabled the development of active solar tracking systems, which became increasingly common [11]. To ensure optimal power output, MPPT systems apply control algorithms that adaptively regulate the PV module's voltage in response to environmental fluctuations, ensuring they consistently operate at or near their maximum power point, even under fluctuating environmental conditions. By minimizing the need for mechanical components, MPPT systems reduce the risk of mechanical failure and associated maintenance costs [14]. Computer algorithms further optimized these movements, enhancing energy output. Over the past few decades, research has expanded to include sophisticated methods such as fuzzy logic, neural networks [15], MPC [16], and RL [17], which provide higher precision and better performance, especially under challenging scenarios like partial shading and rapid environmental changes [18]. Additionally, a reinforcement learning-based maximum power point tracking (RLMPPT) technique for PV arrays has been introduced, surpassing existing MPPT approaches in efficiency and adaptability to environmental variations. The RLMPPT approach achieves a superior efficiency factor compared to traditional MPPT methods, in both simulated and real-world weather conditions, by rapidly adapting to environmental changes and exhibiting minimal learning duration [19].

III. MPPT CONTROLLER ALGORITHMS

Photovoltaic systems demonstrate a nonlinear power-voltage (P-V) characteristic, and the output power is variable and dependent on environmental variables such as solar irradiance, temperature, and/or shading. To maximize energy harvest, PV systems must operate at their Maximum Power Point (MPP), which is a variable dynamic operating point

throughout the day. MPPT is a necessary control mechanism or algorithm that continually adjusts operating voltage and current so that PV system operates at optimal efficiency. MPPT controllers do this by employing several algorithms that are designed to track the MPP in changing environmental conditions.

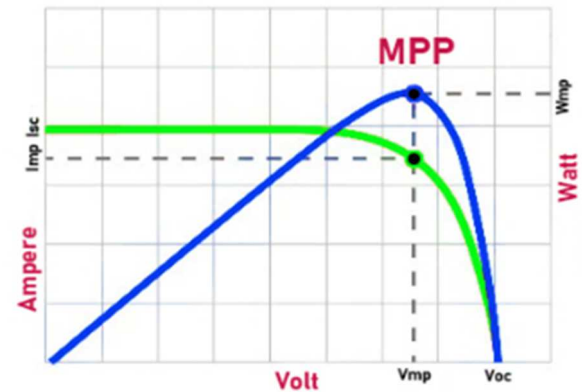


Fig. 1. I-V Plot for MPP

Conventional MPPT strategies employ either fixed rules or iterative search methods. To demonstrate these concepts, Fig. 1 shows the I-V and P-V characteristics of a PV module and marks the MPP — as you can see, this is the best operating point for the solar panel, providing the greatest power output. The P-V curve (blue line in the figure) displays power output versus voltage. Power begins to increase with voltage until it reaches its maximum at the MPP, where voltage times current are at their highest. After the MPP voltage level, an increase in the voltage will cause power output to drop quickly since current falls sharply as well. At MPP, the corresponding voltage (V_{mp}) and current (I_{mp}) define the optimal operating point for maximum energy extraction. Efficient PV systems employ MPPT algorithms to ensure operation at this point, adjusting voltage dynamically to compensate for changes in solar irradiance and temperature. The fundamental principle for adjusting the MPP in PV systems begins with traditional P&O methods, which rely on an iterative process of comparing voltage values to determine the direction of adjustment. Unlike P&O, which relies on basic voltage perturbations, modern MPPT techniques such as MPC and RL utilize mathematical optimization to achieve higher efficiency, improved response time, and better adaptability to real-world conditions, ensuring maximum power extraction even under fluctuating irradiance and partial shading scenarios. The output of any MPPT method is a control signal—typically an optimized duty cycle (D)—that is sent to a DC-DC converter. This duty cycle plays a crucial role in determining the switching behavior of the converter, directly influencing the voltage and current supplied to the load or energy storage system.

Duty cycle (D) signifies the fraction of each switching period the MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor) is ON. Once the MOSFET is ON, the PV panel will provide energy, and that energy is stored in an inductor temporarily. Once the MOSFET turns OFF, i.e. the IGBT, energy stored in the inductor is released to the output so power is supplied to the load in a continuous manner. The MPPT algorithm will adjust the duty cycle to ensure the PV panel operates at maximum power point and extracts as much energy as possible and deliver in along the grid.

A. Perturb and Observe Method

The Perturb and Observe method is the most commonly implemented MPPT technique due to its simplicity and ease of implementation [7]. The Perturb and Observe (P&O) method tracks the Maximum Power Point (MPP) by making small adjustments to the operating voltage of the photovoltaic (PV) system and analyzing the corresponding changes in power output. If an increase in power is observed, the algorithm maintains the same direction of adjustment. Conversely, if the power decreases, the direction of perturbation is reversed. This process iteratively continues until the system converges to the MPP, ensuring optimal power extraction.

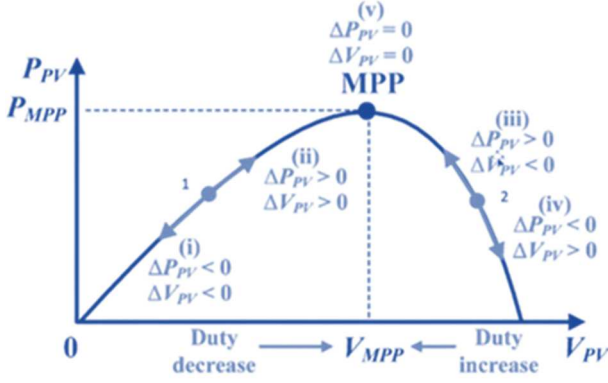


Fig. 2. MPP adjustments using P&O

Fig. 2 represents the P-V characteristic curve of a PV system, illustrating the MPP tracking principle using an MPPT algorithm. The P-V characteristic curve of a PV panel is nonlinear, and the goal of the P&O algorithm is to ensure that the PV array operates at the optimal MPP where the power derivative with respect to voltage is zero presented in (1).

$$\frac{dP}{dV} = 0 \quad (1)$$

P&O operates on the basis of gradually altering the PV voltage (V) and tracking the power (P) change to identify the direction of the subsequent perturbation. The algorithm complies with these requirements:

- If $dP / dV > 0 \rightarrow$ Increasing Voltage (V)
- If $dP / dV < 0 \rightarrow$ Decreasing Voltage (V)

The power P at any instant k can be defined as (2).

$$P(K) = V(K) \cdot I(K) \quad (2)$$

The incremental change in power is given by (3) and (4).

$$\Delta P = P(K) - P(K - 1) \quad (3)$$

$$\Delta V = V(K) - V(K - 1) \quad (4)$$

The P&O method's effectiveness in MPPT is impacted by a number of limitations. Its propensity to oscillate around the MPP as a result of ongoing voltage perturbations, which causes minor power fluctuations, is one of its main disadvantages. Furthermore, P&O struggles to precisely track the MPP in quickly changing environmental conditions due to its slow response time. Its inefficiency under partial shading conditions is another important drawback. This is because it may mistakenly interpret local maxima as the actual MPP, which would lead to less than ideal power extraction from the PV system. P&O is less appropriate for dynamic and complex operating environments because of these factors.

B. Model Predictive Control

MPC utilizes a predictive system model to estimate future responses and adjust control inputs in real time, targeting optimal performance within a defined prediction window based on a receding horizon approach [12] presented in Fig. 3. Initially applied in industrial process control [13], MPC has since been adapted for energy systems, including PV applications. MPC calculates the optimal trajectory of control inputs, such as the orientation of solar panels, by predicting future output and minimizing a cost function (e.g., maximizing energy output or minimizing operational cost).

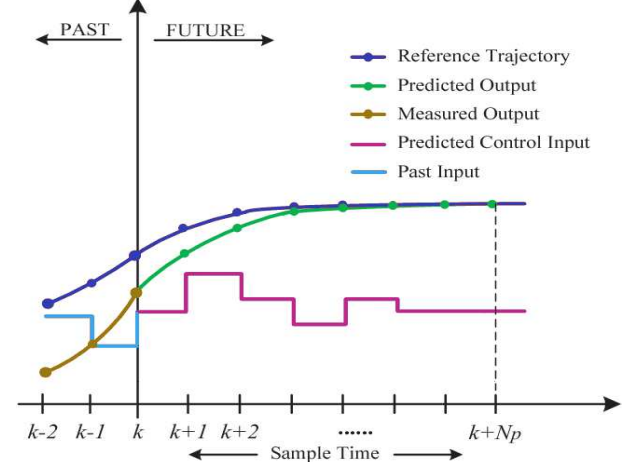


Fig. 3. The model predictive strategy

In PV systems, MPC has been used to optimize both electrical power output through MPPT techniques and mechanical solar tracking. For instance, in PV-battery energy management systems, MPC has demonstrated significant potential in reducing energy costs by optimally scheduling battery charging and discharging based on forecasted solar generation, thus enhancing overall energy utilization and reducing operational costs. In solar tracking, MPC predicts the optimal panel orientation by considering weather forecasts and irradiance data [14].

The fundamental principles underlying Model Predictive Control (MPC) are concisely introduced in here [20]. To predict the system's future performance, a control strategy is applied to ensure that the system achieves the desired outcomes. Model Predictive Control (MPC), a widely used predictive control technique, utilizes a mathematical model of the system to generate these predictions. In the discrete-time state-space form, the model is represented as shown in Equation (5).

$$\begin{aligned} \hat{x}(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned} \quad (5)$$

The system's behavior is described by the state vectors $x(k)$ and $\hat{x}(k+1)$ which represent the state of the system at the current and next time steps, respectively. Similarly, $u(k)$ and $y(k)$ denote the input and output vectors at the current time step. A, B, C, and D correspond to the system dynamics, input effects, output relationships, and disturbances, respectively. To evaluate the system's performance, an objective function J is defined as a function of the system's states and inputs, as shown in Equation (6).

$$J = f(x(k), u(k), \dots, u(k+N)) \quad (6)$$

Model Predictive Control (MPC) operates by predicting the system's performance over a specified future period, known as the **prediction horizon** N , which is a positive integer. At any given instance $k + N$, the system input is represented by the vector $u(k + N)$, while the sequence of previous inputs leading up to this point is also considered in the objective function J , as illustrated in Fig. 3. This figure highlights the system's reference and actual states, along with the discrete inputs applied at various instances, from the past up to $k + N$. As time progresses, the system error diminishes, bringing the actual state closer to the reference state.

Typically, the objective function J accounts for the absolute or squared tracking error between the reference and predicted system output. However, J can also incorporate additional components to meet non-regulatory objectives. In most cases, the objective function is defined as a weighted sum of these different objectives. The primary goal of MPC is to determine the sequence of control inputs that minimizes J , while adhering to system constraints. While deriving a general solution to this optimization problem is complex, a more straightforward approach involves evaluating the value of J for all possible inputs during each sampling period and selecting the input that yields the lowest J for the next period. This process generates a sequence of optimal input signals, but the controller applies only the first input in the sequence at each step. After applying this input, the system model and the objective function are updated using the most recent data, effectively shifting the prediction horizon forward [20].

Despite its many advantages, MPC has certain limitations. Its performance is highly dependent on the accuracy of the system model, which can be challenging to maintain in real-world applications where environmental conditions fluctuate. Additionally, MPC often demands considerable computational resources, which can restrict its use in real-time applications.

C. Reinforcement Learning

Reinforcement Learning (RL) focuses on learning to make sequential decisions via an agent who interacts with its environment and aims to optimize long-term reward in Fig. 4. Unlike conventional control approaches, RL does not use a model of the environment; instead, it learns from interactions with the environment over time [15]. Early RL applications were seen in robotics and gaming, but with advances in computational power, RL has been applied to energy management and optimization [21]. In PV systems, RL has been particularly useful in optimizing MPPT without needing a detailed model of the PV system. By leveraging its ability to learn optimal control policies based on interactions with the environment [18], RL-based MPPT techniques adapt to changes in sunlight intensity and temperature faster than conventional methods such as P&O or incremental conductance. Additionally, RL can be integrated with solar tracking systems to dynamically adjust panel orientation, learning from past performance to improve future energy capture [22].

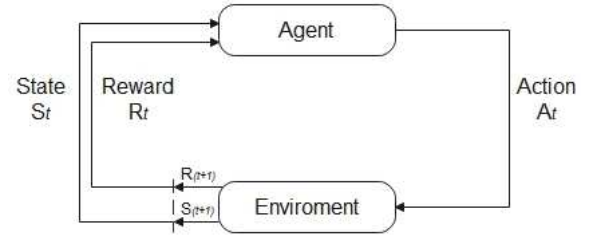


Fig. 4. Reinforcement control methodology

Unlike traditional methods, RL-based tracking systems can adapt to unpredictable and complex environmental changes without requiring a predefined model of the PV system [23].

The application of RL has been particularly beneficial in developing advanced control schemes for PV systems in dynamically changing environments. For example, Deep Reinforcement Learning (DRL) algorithms like Deep Deterministic Policy Gradient (DDPG) have outperformed traditional methods by providing more accurate MPPT under varying irradiance and temperature conditions. Research has demonstrated that a DDPG RL agent can enhance synergetic and sliding mode controllers, ensuring parametric robustness and maintaining system performance despite significant load variations. This integration of RL facilitates adaptive learning capabilities and allows for more precise control strategies that adjust in real time to changing conditions [24]. To determine the optimal policy, several algorithms leverage the value function $V^\pi(s)$, which defines how good it is for the agent to reach a given state [25]. The expected return associated with following a policy π from a given state s is denoted by the state-value function. Furthermore, some strategies depend on the action-value function, represented by denoted as $Q^\pi(s, a)$, which shows the expected return of taking this action a in the current state s under a policy π . The $V^\pi(s)$ and $Q^\pi(s, a)$ functions are calculated from (7) and (8) as below [15].

$$V^\pi(s_t) = E[R_t | s_t = s] = E[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} | s_t = s] \quad (7)$$

$$Q^\pi(s_t, a_t) = E[R_t | s_t = s, a_t = a] \\ = E[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} | s_t = s, a_t = a] \quad (8)$$

Q-Learning is a popular model-free, off-policy reinforcement learning (RL) algorithm that has gained widespread use in various fields. At its core, Q-Learning works by estimating the Q-value for each state-action pair (s, a) , which represents the expected future rewards when taking a certain action a from a state s , then following an optimal policy. This estimation process follows an iterative update derived from the Bellman equation, expressed by (9):

$$Q^\pi(s_t, a_t) = E[r_{t+1} + \gamma Q^\pi(s_{t+1}, a_{t+1}) | s_t, a_t] \quad (9)$$

An optimal policy π^* achieves the maximum cumulative reward over the long term. Currently, the action-value function and best value function are provided by

$$\pi^* = \underset{\pi}{\operatorname{argmax}} V^\pi(s) \quad (10)$$

$$V^*(s) = \max_{\pi} V^\pi(s) \quad (11)$$

$$Q^*(s, a) = \max_{\pi} Q^\pi(s, a) \quad (12)$$

The primary limitation of RL is the long training time required to learn an optimal policy, especially in systems where real-time decisions are crucial. Additionally, RL can struggle with

stability and performance when applied in highly stochastic environments, such as solar energy systems where weather and irradiance change unpredictably.

IV. FINDINGS AND RESULTS

The simulation analysis takes place on a computer simulation platform for an established string system that is built from a PV panel, a necessary MPPT controller, a buck converter (DC-DC converter), and a connection to a battery to assess the effectiveness of various MPPT modes of operation for power output. The testing conditions discussed in this work are two cases, case-1 for ideal conditions where the irradiation is 1000 W/m^2 with 25°C of temperature ensuring optimal energy generation, whereas case-2 is under harsh conditions with 500 W/m^2 and 45°C simulating real-world challenges such as partial shading and temperature fluctuations. The system performances are discussed with regard to the three methods of MPC, P&O and RL for comparison.

A. Case-1 Ideal Conditions

Fig. 5 illustrates the power output of a PV system under standard test conditions (1000 W/m^2 irradiance and 25°C temperature) using three different control algorithms: RL, MPC, and P&O. The initial response of RL exhibits a brief dip below zero before rapidly increasing, while MPC also experiences an initial overshoot but stabilizes more smoothly. P&O, on the other hand, demonstrates the highest overshoot and transient oscillations before reaching a steady state. In terms of oscillations and settling time, RL and MPC stabilize faster, with MPC exhibiting fewer fluctuations compared to P&O. Although all three methods ultimately converge to a similar power output, P&O's prolonged oscillations suggest less effective optimization in dynamically tracking the maximum power point. RL appears promising but shows an initial negative dip, likely due to its adaptive learning process adjusting to system dynamics. In general, MPC and RL achieve faster settling times and reduced overshoot during transient states, outperforming the P&O method.

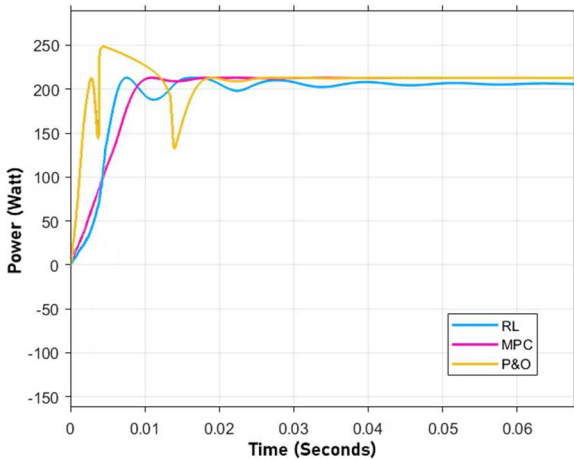


Fig. 5. Comparison among P&O, MPC, and RL (Irradiance = 1000 W/m^2 , Temperature = 25°C)

B. Case-2 Harsh Conditions

The power output of a PV system using the same three control algorithms at higher temperatures (45°C) and lower irradiance (500 W/m^2) is shown in Fig. 6. As would be expected given the decreased irradiance and elevated temperature, which have a detrimental effect on PV efficiency, the power output is

lower under these circumstances than it was in the prior scenario. In comparison to P&O, the RL and MPC algorithms stabilize at a higher power level, suggesting superior tracking of the maximum power point. Whereas RL oscillates at first before rapidly settling, MPC maintains a comparatively smooth response with few oscillations. P&O, on the other hand, shows notable oscillations and a lower steady-state power output, underscoring its shortcomings in efficiently managing dynamic environmental conditions. Overall, MPC and RL perform better than P&O in both transient and steady-state scenarios; RL exhibits adaptive behavior, while MPC provides the smoothest response. Table I compares the power generation performances..

TABLE I. POWER PERFORMANCE COMPARISON

Condition	Algorithm	Oscillations	Settling Time	Steady-State
1000 W/m^2 , 25°C	RL	Moderate	Fast	High
	MPC	Minimal	Fast	High
	P&O	High	Slow	High
500 W/m^2 , 45°C	RL	Minimal	Fast	Highest
	MPC	Moderate	Fast	Moderate
	P&O	High	Slow	Low

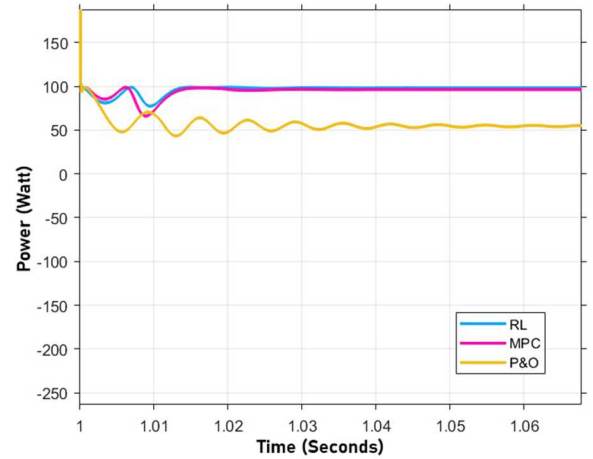


Fig. 6. Comparison among P&O, MPC, and RL (Irradiance = 500 W/m^2 , Temperature = 45°C)

The table shows that the MPC is the best in Case-1 (1000 W/m^2 , 25°C), providing the smoothest response with minimal oscillations but struggles slightly under lower irradiance. It is observed that RL is the best in Case-2 (500 W/m^2 , 45°C), achieving higher steady-state power and settling faster than the other methods. Finally the P&O is consistently the weakest performer, struggling with high oscillations and slow convergence in both cases.

Conclusion

This study examined how MPC and RL compare to the conventional, widely used P&O method for MPPT in PV systems. The analysis reveals key distinctions between their adaptability, computational needs, and operating principles. MPC is appropriate for structured environments with clearly defined system dynamics because of its model-based predictive approach, which guarantees methodical decision-making and constraint handling. However, RL is a promising option for applications with uncertain or changing conditions because it offers a data-driven solution that adjusts to environmental variations. The trade-offs between adaptability, convergence speed, and computational

complexity suggest that particular system requirements should be taken into consideration when choosing an MPPT method. To improve MPPT performance, future research may look into hybrid strategies that combine adaptive learning and predictive control.

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